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A Physiologically Based Pharmacokinetic Model to Predict Potential Drug–Drug Interactions of TPN171, a Novel Phosphodiesterase Type 5 Inhibitor

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ABSTRACT

TPN171 is a potent phosphodiesterase type 5 inhibitor used to treat pulmonary arterial hypertension and erectile dysfunction, primarily metabolized by CYP3A4. This study aimed to evaluate its drug–drug interaction (DDI) potential and determine the optimal dosing when co-administered with CYP3A4 modulators. A physiologically based pharmacokinetic (PBPK) model was developed and validated using clinical DDI data for itraconazole (strong CYP3A4 inhibitor) and rifampin (strong CYP3A4 inducer). The model was then applied to predict DDIs with moderate (diltiazem, fluconazole) and mild (fluvoxamine) inhibitors, as well as moderate (efavirenz) and mild (zanubrutinib) inducers. Predicted AUC_{0-t} ratios aligned with observed data: itraconazole (2.42-fold predicted vs. 2.67-fold observed) and rifampicin (0.14-fold predicted vs. 0.096-fold observed). For other modulators, geometric mean AUC_{0-t} ratios were 2.40 with diltiazem, 3.27 with fluconazole (400 mg), 2.07 with fluconazole (100 mg), 1.16 with fluvoxamine, 0.53 with efavirenz, and 0.70 with zanubrutinib. These simulations indicate that strong and moderate CYP3A4 inhibitors and inducers significantly alter TPN171 exposure, whereas weak modulators have no clinically meaningful impact. These findings provide a scientific basis for dose recommendations when TPN171 is used with CYP3A4 modulators.

1 | Introduction

TPN171 is a novel, potent, and selective inhibitor of phosphodiesterase type 5 (PDE5) under development for pulmonary arterial hypertension (PAH) and erectile dysfunction (ED) [1–3]. PDE5 is abundantly expressed in pulmonary vascular smooth muscle and penile corpus cavernosum, where it functions as the primary enzyme responsible for hydrolyzing cyclic guanosine monophosphate (cGMP) [4]. Inhibition of PDE5 elevates intracellular cGMP levels, thereby potentiating nitric oxide (NO)-mediated vasodilation and smooth muscle relaxation, which improves pulmonary hemodynamics and facilitates penile erection [5, 6].

Clinical pharmacokinetic (PK) studies in healthy subjects have demonstrated rapid absorption of TPN171, with a median T_{max} of 0.9–1.3 h, linear PK across the 5–30 mg dose range, and a terminal half-life of 8.0–10.9 h supporting once-daily dosing for PAH [7]. Following oral administration of [¹⁴C]-labeled TPN171 (10 mg, 100 μ Ci), recovery of total radioactivity was nearly complete, with 46.61% excreted in urine and 48.60% in feces. Unchanged drug accounted for 11.48% and 7.08% of the administered dose in urine and feces, respectively [8, 9]. An 8-week phase 2 trial and a 12-week phase 3 pivotal trial have established the efficacy and safety of on-demand TPN171 for ED in Chinese men. Effective doses ranged from 5 to 20 mg in phase 2 (manuscript under review) and from 2.5 to 10 mg in phase 3 [10].

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Study Highlights

- What is the current knowledge on the topic?
 - TPN171 is expected to exhibit clinically significant drug interactions with cytochrome P450 3A4 (CYP3A4) modulators, as demonstrated in clinical studies involving itraconazole and rifampicin.
- What question did this study address?
 - To develop and qualify a PBPK model of TPN171 for the assessment of CYP3A4-mediated drug-drug interactions.
- What does this study add to our knowledge?
 - The PBPK simulations indicated that strong and moderate CYP3A4 inhibitors as well as strong and moderate CYP3A4 inducers significantly alter TPN171 exposure, whereas weak CYP3A4 inhibitors and weak CYP3A4 inducers exhibited no clinically relevant impact.
- How might this change drug discovery, development, and/or therapeutics?
 - These findings provide a scientific basis for formulating dose recommendations when TPN171 is used in combination with CYP3A4 inhibitors or inducers.

Drug–drug interactions (DDIs) can be mediated not only by the modulation of cytochrome P450 (CYP) enzymes, but also via inhibition or induction of membrane drug transporters. Key transporters with established clinical relevance—including P-glycoprotein (P-gp), breast cancer resistance protein (BCRP), organic anion transporting polypeptides (OATP1B1/1B3), organic anion transporters (OAT1/3), organic cation transporter 2 (OCT2), and multidrug and toxin extrusion proteins (MATE1/2K)—are now routinely evaluated during drug development to inform clinical DDI strategies and regulatory labeling [11]. In the present study, a comprehensive transporter evaluation was conducted to characterize the interaction profile of TPN171. The results demonstrated that TPN171 is not a substrate of CYP isoforms other than CYP3A4, nor is it a substrate of BCRP, OATP1B1, OATP1B3, OAT1, OAT3, OCT2, MATE1, or MATE2K. However, TPN171 exhibited concentration-dependent inhibition of several renal transporters, with IC_{50} values of 11.8 μ M for OCT2, 1.57 μ M for MATE1, and 0.367 μ M for MATE2-K. No significant inhibition was observed against BCRP, OATP1B1, OATP1B3, OAT1, or OAT3 ($IC_{50} > 50 \mu$ M).

A dedicated clinical DDI study (NCT06090123) confirmed that potent CYP3A4 modulators substantially alter TPN171 exposure. Coadministration with itraconazole (200 mg once daily) increased TPN171 AUC_{0-t} and C_{max} by 167% and 76%, respectively, whereas rifampicin (600 mg once daily) decreased these parameters by 90% and 75% [12].

Physiologically based pharmacokinetic (PBPK) modeling integrates drug properties and physiological parameters to simulate absorption, distribution, metabolism, and excretion (ADME) processes and predict DDIs involving both enzymes and transporters [13]. This approach is increasingly accepted

by regulatory agencies to inform clinical strategy and support labeling decisions [14, 15]. GastroPlus (Simulations Plus Inc.) is a platform and database for PBPK modeling and simulation that predicts PK profiles and DDIs [16, 17]. In this study, we developed and validated a whole-body PBPK model for TPN171. The model incorporates comprehensive in vitro data on CYP3A4 metabolism and transporter interactions to quantitatively predict the magnitude of DDIs across a spectrum of CYP3A4 modulators. These simulations provide a rational basis for dose adjustment recommendations when TPN171 is co-administered with CYP3A4 inhibitors or inducers.

2 | Methods

2.1 | Overview of Modeling Strategy

The PBPK model for TPN171 was developed using experimental physicochemical, in vitro, and clinical data obtained from healthy volunteers (HVs). The model was validated against data from several clinical studies, including DDI studies, single ascending dose (SAD), multiple ascending dose (MAD), food effect (FE), and PK studies in the elderly (Table S1). Subsequently, the PBPK model was applied to predict the potential impact of co-administering CYP3A4 inhibitors or inducers on the PK of TPN171 in HVs. The overall modeling strategy is illustrated in Figure 1.

2.2 | PBPK Model Development

A PBPK model was developed to simulate the PK of TPN171 in HVs using GastroPlus (Version 9.8.3002, Simulations Plus Inc.). The physicochemical properties of TPN171 were determined through in vitro assays, while its human ADME and mass balance data were obtained from dedicated clinical studies (Table 1). Demographic and administration information was derived from clinical DDI studies [12]. PK data from the TPN171-alone arms of these DDI studies following single oral doses of 10 mg or 20 mg were used to optimize the PBPK model.

The Advanced Compartmental Absorption and Transit (ACAT) model implemented in GastroPlus was used to describe the dissolution and absorption of orally administered TPN171 [18]. The effective permeability (P_{eff}) value of TPN171 was estimated based on in vitro permeability data. A correlation was established using the in vitro permeabilities and in vivo P_{eff} values of reference compounds (digoxin and propranolol) in a Caco-2 assay [17]. P-gp-mediated efflux kinetics were incorporated into the ACAT model to simulate intestinal efflux during TPN171 absorption. The predicted fraction absorbed (F_a) was compared with the amount of unchanged TPN171 recovered in feces from a human mass balance study [9]. Additionally, CYP3A4 kinetics were integrated into the ACAT model to simulate gut first-pass metabolism of TPN171.

2.3 | PBPK Model Validation

The TPN171 PBPK model was validated using pharmacokinetic data from clinical DDI studies in which TPN171 was

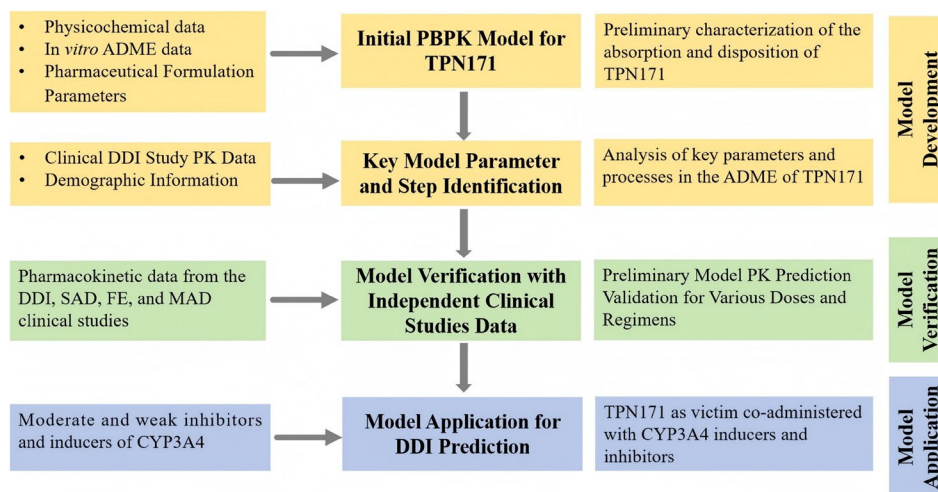


FIGURE 1 | Flowchart for overall modeling strategy. ADME, Absorption, Distribution, Metabolism, Excretion; PK, pharmacokinetics; DDI, drug–drug interaction; SAD, single ascending dose; MAD, multiple ascending dose; FE, food effect; PBPK, physiologically based pharmacokinetic.

co-administered with itraconazole or rifampin [12]. To assess the model's ability to capture inter-individual variability, population simulations were performed across five virtual trials, each comprising 12 healthy volunteers under co-administration with itraconazole or rifampin.

For model validation, the primary evaluation metrics were the $C_{\max}$ ratio ($C_{\max}R$) and the AUC ratio (AUCR), defined as the ratio of TPN171 $C_{\max}$ or AUC when co-administered with a CYP3A4 modulator to that when administered alone. The prediction error for these parameters was assessed by comparing the simulated values with the observed values from clinical trials, using the Guest Equation (1) as the acceptance criterion [19–21].

$$Limit = \frac{1 + 2(R_{obs} - 1)}{R_{obs}} \quad (1)$$

where R_{obs} represents the clinically observed $C_{\max}R$ or AUCR, with $R_{obs} > 1$. For induction effects, R_{obs} is defined as the reciprocal of the observed $C_{\max}R$ or AUCR. The acceptable upper limit for model prediction error is $R_{obs} \times Limit$, and the acceptable lower limit is $R_{obs}/Limit$.

The TPN171 PBPK model was further validated against clinical data from SAD and MAD studies across the 5–30 mg dose range, FE studies with 10 mg TPN171 administered under fasted and fed conditions, and pharmacokinetic studies in elderly subjects [7, 22]. Following oral administration of TPN171 alone in DDI, SAD, and MAD clinical studies, the prediction error for key pharmacokinetic parameters was estimated by comparing simulated values with observed values from clinical trials [7, 18], using Equation (2). A trial simulation was considered successful if the prediction error fell within the 0.8–1.25 range [23, 24].

$$Prediction\ error = T_{predicted} / R_{observed} \quad (2)$$

where $T_{predicted}$ represents the simulated geometric mean $C_{\max}$ and AUC for TPN171 exposure from virtual clinical trials, and

$R_{observed}$ denotes the corresponding observed geometric mean values obtained from actual clinical trials.

2.4 | PBPK Model Application

The validated PBPK model was applied to predict and evaluate changes in TPN171 pharmacokinetics when co-administered with clinically relevant moderate or weak CYP3A4 modulators. Based on the published DDI modeling approaches for selected marketed drugs [25–27], diltiazem and fluconazole were classified as moderate CYP3A4 inhibitors, fluvoxamine as a weak CYP3A4 inhibitor, efavirenz as a moderate CYP3A4 inducer, and zanubrutinib as a weak CYP3A4 inducer. A dedicated DDI simulation strategy was developed by integrating commonly used dosing regimens from DDI studies with the established clinical DDI trial design for TPN171. The corresponding results are summarized in Table 2.

2.5 | PBPK Models of Other Compounds

The DDI models used in this study (with the exception of zanubrutinib) were derived either from the latest validated model files provided by the software developer or from the most recent version of the built-in DDI standard library. These models have undergone thorough validation and have been evaluated for their DDI prediction performance. They have also been employed in regulatory submissions or academic research involving other DDI models, supporting the reliability of DDI predictions when TPN171 is combined with other drugs.

3 | Results

3.1 | PBPK Model Development

The PBPK model was developed using pharmacokinetic data from healthy male volunteers who received single oral doses of 10 mg or 20 mg TPN171 under fasted conditions in clinical DDI

TABLE 1 | Input physicochemical and pharmacokinetic parameters for the TPN171 PBPK modeling.

Parameter	Value	Source
Molecular weight (g/mol)	478.03	Experimental data
log P	2.45	Experimental data
pKa	Base, 8.3	Experimental data
Aqueous solubility (mg/mL)	458.0 at pH 7.0	Experimental data The Solubility Factor (SF) was fitted to a value of 21.5 by integrating the experimentally determined in vitro values, the pKa, and the software-calculated values
Solubilization ratio	5302.7	Predicted by GastroPlus
Precipitation time (s)	900	Default by GastroPlus
Diff. Coeff. (cm ² /s)	0.5983*10 ⁻⁵	Estimation by GastroPlus combining molecular weight
Particle density (g/mL)	0.26	Experimental data
Particle size (μm)	D10: 2.60 D50: 8.61 D90: 36.32	Experimental data A log-normal particle size distribution curve was fitted in the model based on the experimentally observed values
Human permeability (cm/s)	Inhibited P-gp: 1.9856*10 ⁻⁴	Converted from in vitro experimental data
R _{bp}	0.74	Experimental data
f _{up}	6.4%	Experimental data
f _{u, ent}	100%	Default by GastroPlus
Volume of distribution (L)	~90	Predicted by GastroPlus human PBPK model
Elimination half time (h)	8.02~10.88	Prediction based on clinical data
fmCYP3A4	0.658	Prediction based on clinical data
CYP3A4 K _m (μM)	36.1	Experimental data
CYP3A4 V _{max} (pmol/min/pmol)	12.56	Experimental data
CL _{H, linear} (L/h)	~0.233	Predicted by GastroPlus
P-gp K _m (mg/L)	3.02	Assume the same with that of CYP3A4
P-gp V _{max} (pmol/min/mg protein)	1.1	Optimized value according the unchanged drug in feces and urine
CL _{R, linear} (L/h)	~1.65	Estimation by GastroPlus based on in vivo data
CL _{sys}	~6.7 L/h	Estimated using the equation t _{1/2} = 0.693 × V _{ss} /CL.
First pass extraction	Gut:~20% Liver:~15%	Calculated by GastroPlus based on In vivo data

Abbreviations: Log P, octanol water partition coefficient; pKa, acid-base dissociation constant; SR, solubilization ratio; Diff. Coeff, diffusion coefficient in aqueous media; R_{bp}, blood to plasma concentration ratio; f_{up}, unbound fraction in plasma protein; f_{u, ent}, unbound fraction in enterocytes; V_{ss}, volume of distribution; fm, fraction of metabolism; K_m, Michaelis-Menten Constant; V_{max}, maximum metabolic rate; f_{u, CYP}, unbound fraction in rhCYP3A4; CL_{H, linear}, linear hepatic clearance mediated by non-CYP3A4, CL_{R, linear}, linear renal clearance mediated by non-P-gp; CL_{sys}, Total system clearance.

studies. Model predictions were consistent with the observed data (Figure S1). The DDI parameters of this perpetrator were listed in Table S4. The predicted C_{max} and AUC_{0-t} values showed good agreement with observed values, with all simulated-to-observed ratios falling within the range of 0.91 to 1.04 (Table S2).

Virtual trials were conducted using five simulation groups, each consisting of 12 virtual subjects receiving a single oral dose of 10 mg or 20 mg TPN171. The predicted plasma concentration-time profiles were largely within the 90% confidence intervals of the observed data (Figure S2). The predicted C_{max}, AUC_{0-t}, and AUC_{0-∞} values were consistent with those observed in the

clinical DDI study, demonstrating satisfactory model stability and predictive performance for inter-individual variability (Table S2).

3.2 | PBPK Model Validation

The PBPK model was validated using clinically observed DDI data from healthy volunteers who received a single oral dose of 10 mg TPN171 co-administered with 200 mg itraconazole, or a single oral dose of 20 mg TPN171 co-administered with 600 mg rifampin [12]. The model-predicted plasma

concentration-time curves for TPN171 alone and in combination showed close agreement with the observed clinical data (Table 3, Figure S3).

Model simulations indicated that concomitant administration of itraconazole increased TPN171 C_{max} by 1.68-fold and AUC_{0-t} by 2.19-fold relative to TPN171 alone. These findings were further corroborated by population simulations, which yielded geometric mean ratios of 1.79 for C_{max} and 2.42 for AUC_{0-t} under co-administration conditions (Table 3, Figure 2).

In contrast, when TPN171 was co-administered with rifampicin, the model projected reduced exposure, with C_{max} and AUC_{0-t} declining to 0.33 and 0.14 times those observed with TPN171 alone, respectively. Consistent with this trend, population simulations produced geometric mean ratios of 0.31 for C_{max} and 0.14 for AUC_{0-t} in the rifampicin DDI scenario (Table 3, Figure 2).

When evaluated against the DDI prediction error acceptance criteria proposed by Guest et al., the model-predicted changes in TPN171 pharmacokinetic parameters following co-administration with itraconazole or rifampicin fell within the acceptable margin of prediction error, demonstrating the robustness and accuracy of the established PBPK model (Table S3).

The PBPK model was further validated against clinical data from SAD studies (5–30 mg), MAD studies (5–10 mg), FE studies, and pharmacokinetic studies in elderly subjects (Figures S4–S7). These findings demonstrate that the model-predicted pharmacokinetic profiles across various clinical scenarios generally aligned with the observed data, with most PK parameter differences falling within the predefined acceptance criteria. The exception was the C_{max} prediction for the 5 mg single-dose group, which fell slightly below the lower bound of the acceptable range—a deviation that may be attributed to the relatively small sample size ($n = 3$) in this dosing cohort (Table S4).

TABLE 2 | TPN171 clinical study data used in PBPK model development and verification.

Inhibitor/Inducer	Dose (mg)	Dosage form	Treatment (days)	Dose regimen
Strong CYP3A4 inhibitors				
Itraconazole*	200	Capsule	4	q.d.
Moderate CYP3A4 inhibitors				
Diltiazem	60	Capsule	8	t.i.d.
Fluconazole	100	Capsule	8	q.d.
Fluconazole	400	Capsule	8	q.d.
Mild CYP3A4 inhibitors				
Fluvoxamine	50	Tablet	8	q.d.
Strong CYP3A4 inducers				
Rifampicin*	600	Capsule	8	q.d.
Moderate CYP3A4 inducer				
Efavirenz	400	Tablet	15	q.d.
Mild CYP3A4 inducer				
Zanubrutinib	160	Capsule	15	b.i.d.

*The clinical PK study have been completed, q.d., once daily; b.i.d., twice daily; t.i.d., every 8 h.

TABLE 3 | PBPK model-predicted geometric mean ratios of C_{max} and AUC for TPN171 as a DDI victim compared with observed data.

	C_{max} R[ng/mL]			AUC_{0-t} R[ng*h/mL]		
	Simulated	Observed ^a	Ratio ^b	Simulated	Observed ^a	Ratio ^b
TPN171 10 mg + itraconazole 200 mg	1.68	1.74	0.97	2.19	2.72	0.81
TPN171 20 mg+ rifampin 600 mg	0.33	0.27	1.22	0.14	0.092	1.49
TPN171 10 mg + itraconazole 200 mg-PS	1.70 (1.63–1.78)	1.76 (1.60–1.93)	0.97	2.42 (2.31–2.54)	2.67 (2.48–2.88)	0.90
TPN171 20 mg+rifampin 600 mg-PS	0.31 (0.28–0.35)	0.25 (0.22–0.30)	1.24	0.14 (0.13–0.17)	0.096 (0.088–0.10)	1.49

^aThe observed values were calculated from GastroPlus based on the plasma concentration-time profiles of clinical study.

^bRatio for simulated values/observed values.

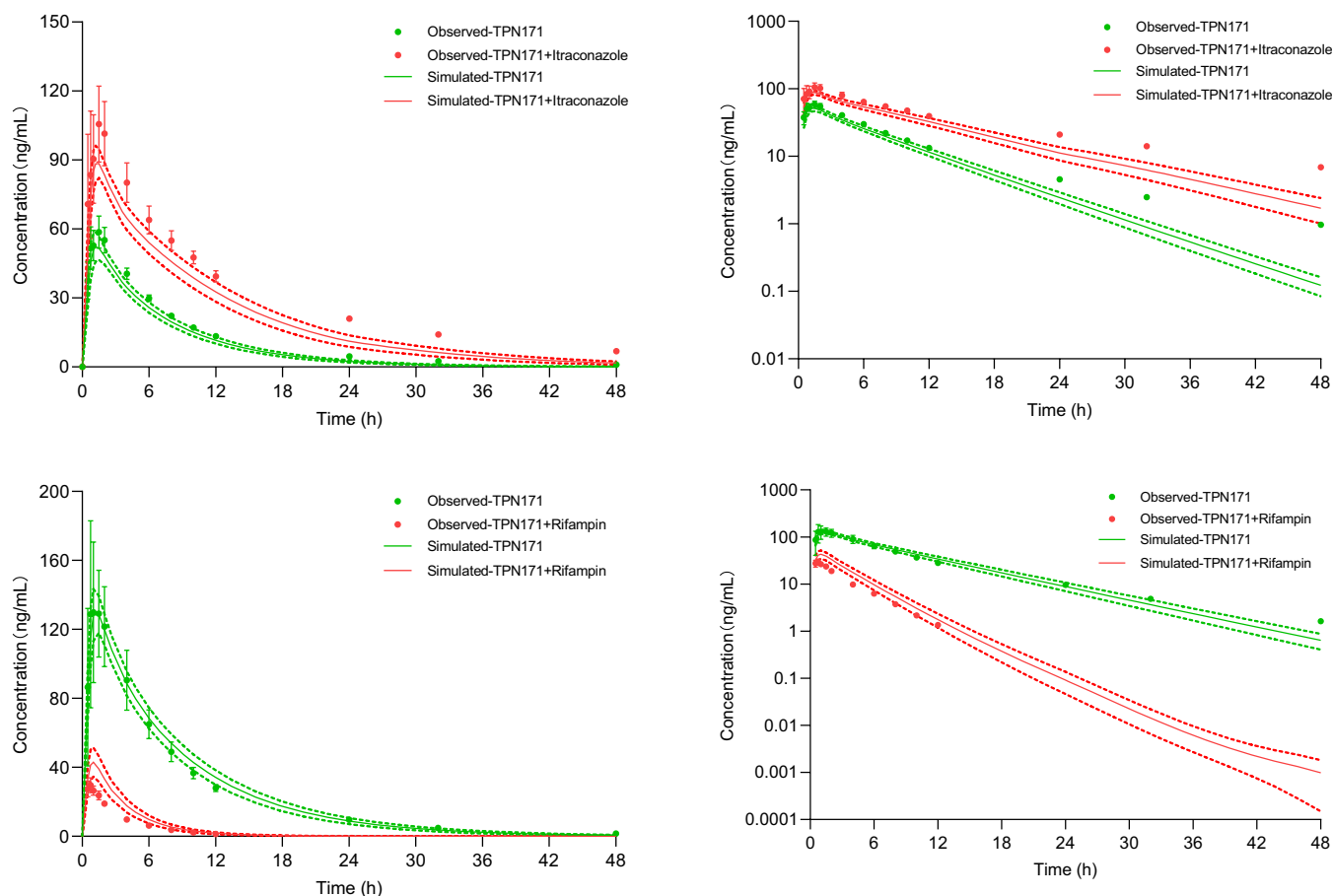


FIGURE 2 | Population simulation of TPN171 for plasma concentration–time profiles. After single oral dosing of TPN171 alone and co-administration with itraconazole or rifampin in healthy volunteers. The solid line (green: TPN171 alone; red: Co-administration with itraconazole or rifampin) and dash dots represent predicted mean plasma concentration and its 95% confidence intervals, respectively (12 virtual individuals), and the solid dots are the observed data from clinical studies ($n = 12$).

TABLE 4 | Summary of all drug–drug interaction prediction.

Substrates	Perpetrators ^a	$C_{\max}R^c$	$AUC_{0-\infty}R$	$AUC_{0-t}R$
TPN171 (10 mg, single)	Itraconazole ^b (200 mg, q.d.)	1.76 (1.60–1.93)	2.86 (2.62–3.12)	2.67 (2.48–2.88)
	Itraconazole (200 mg, q.d.)	1.70 (1.63–1.78)	2.47 (2.34–2.59)	2.42 (2.31–2.54)
	Diltiazem (60 mg, t.i.d.)	1.77 (1.65–1.93)	2.40 (2.09–2.92)	2.40 (2.09–2.92)
	Fluconazole (400 mg, q.d.)	1.47 (1.43–1.50)	3.60 (3.33–3.89)	3.27 (3.03–3.52)
	Fluconazole (100 mg, q.d.)	1.34 (1.31–1.36)	2.11 (2.00–2.23)	2.07 (1.96–2.17)
	Fluvoxamine (50 mg, q.d.)	1.10 (1.09–1.11)	1.16 (1.14–1.18)	1.16 (1.14–1.18)
	Zanubrutinib (160 mg, b.i.d.)	0.79 (0.76–0.82)	0.70 (0.68–0.73)	0.70 (0.68–0.73)
	Efavirenz (400 mg, q.d.)	0.79 (0.73–0.84)	0.52 (0.46–0.60)	0.53 (0.46–0.61)
TPN171 (20 mg, single)	Rifampin (600 mg, q.d.)	0.31 (0.28–0.35)	0.143 (0.124–0.166)	0.144 (0.125–0.166)
	Rifampin ^b (600 mg, q.d.)	0.25 (0.22–0.30)	0.099 (0.091–0.107)	0.096 (0.088–0.104)

^aq.d., once a day; t.i.d., three times a day.

^bobserved values from drug–drug interaction clinical studies.

^cRatios for TPN171 alone values/TPN171 co-administration with CYP3A4 perpetrators values.

3.3 | PBPK Model Application

The simulated effects of moderate and weak CYP3A4 inhibitors on TPN171 pharmacokinetics are summarized in Table 4

and illustrated in Figure 3. Model-based simulations yielded the following geometric mean ratios for TPN171 $C_{\max}$ and AUC_{0-t} : 1.77 and 2.40 with the moderate inhibitor diltiazem; 1.47 and 3.27 with fluconazole 400 mg (moderate inhibitor); 1.34 and 2.07

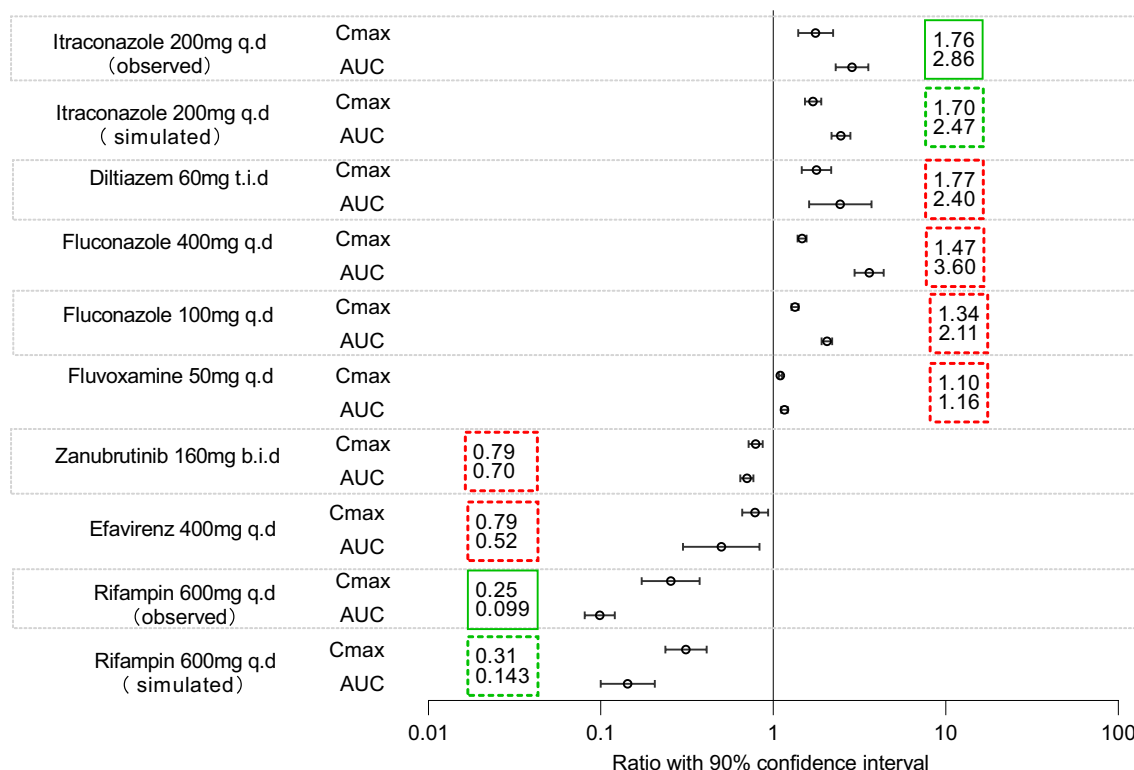


FIGURE 3 | Summary of DDI predictions using the TPN171 PBPK models.

with fluconazole 100 mg (moderate inhibitor); and 1.10 and 1.16 with the weak inhibitor fluvoxamine.

For CYP3A4 inducers, simulations showed geometric mean C_{max} and AUC_{0-t} ratios of 0.79 and 0.53 when TPN171 was co-administered with the moderate inducer efavirenz, and 0.79 and 0.70 with the weak inducer zidovudine.

4 | Discussion

This study successfully developed and validated a human PBPK model for TPN171 tablets by integrating its theoretical and biopharmaceutical parameters, in vitro formulation data, and preclinical or clinical pharmacokinetic data, utilizing the ACAT and PBPK models within the GastroPlus software. The verified model was subsequently applied to assess DDIs between TPN171 and moderate or weak CYP3A4 perpetrators, providing evidence-based recommendations for dosing regimen adjustments during co-administration.

A PBPK model for TPN171 was developed using data from single oral doses of 10mg and 20mg in humans. The model was qualified by comparing its predictions with the observed PK data from Day 1 of the clinical DDI study [12]. The predicted PK profiles generally aligned well with the observed mean data, falling within the error ranges of the observed values. Key predicted PK parameters (C_{max} , AUC_{0-t}) were within the 0.80- to 1.25-fold range of the observed values. Furthermore, when inter-individual variability was incorporated, the simulated population PK variability ranges largely encompassed the individually

observed clinical PK data points. The model-predicted amounts of unabsorbed drug and unchanged drug in urine were also consistent with the mass balance results [9].

Based on PBPK modeling and analysis, TPN171 exhibited rapid and complete dissolution following oral administration in humans, followed by rapid permeability and absorption into enterocytes, consistent with the short T_{max} observed in the PK profiles [7]. A portion of TPN171 at the apical side of enterocytes was effluxed back into the intestinal lumen via P-gp, resulting in incomplete absorption (Fa~93%), which aligns with the detection of 7.08% unchanged drug in feces from the human mass balance study [9]. Within enterocytes, approximately 20% of the absorbed TPN171 underwent gut first-pass metabolism mediated by CYP3A4, resulting in about 75% of the administered dose reaching the hepatic portal vein.

TPN171 demonstrated moderate plasma clearance, with an elimination half-life of approximately 9.2h. The PBPK model predicted systemic clearance comprised CYP3A4-mediated hepatic metabolism (~65%), metabolism by other enzyme subtypes (~3.5%), P-gp-mediated active renal secretion (~5.1%), and non-P-gp-mediated renal clearance (~25.6%). An additional~25% hepatic first-pass metabolism further reduced systemic availability, resulting in a predicted oral bioavailability of approximately 65%. The PBPK model suggested extensive tissue distribution for TPN171, with a predicted volume of distribution at steady state (V_{ss}) of approximately 1.4L/kg. Parameter sensitivity analysis indicated that V_{ss} was relatively insensitive to large variations in the fraction unbound in plasma (f_{up}), potentially because TPN171 primarily exists in a cationic form at plasma

pH 7.4. The model also predicted that approximately 11% of the oral dose was excreted unchanged in urine, consistent with the range observed in clinical studies (10%-20% of dose). The overall agreement between the PBPK model predictions (both PK profiles and parameters) and the observed data, coupled with the consistency of the model-described ADME processes with existing *in vitro* and *in vivo* findings [1], supports the validity and accuracy of the initial human PBPK model for TPN171.

During model verification for the single-dose, fasted 5 mg TPN171 group, the predicted $C_{\max}$ was slightly lower than the observed value (predicted/observed ratio = 0.73). This underprediction is likely attributable to the limited sample size ($n=3$) in this dose cohort, suggesting that the observed mean PK profile may not robustly represent the typical pharmacokinetic characteristics at this dose level [7]. Additionally, for multiple-dosing simulations, GastroPlus calculates the total AUC from the first to the last dose but cannot compute the AUC specifically for the final dosing interval. This limitation prevented a direct accuracy comparison of PK parameters with the observed plasma concentration-time data points from days 9 to 11. Consequently, for the multiple-dosing model validation, the current work focused on comparing the prediction error for PK parameters after the first dose and visually assessing the concordance between predicted and observed PK profiles.

When validating the model against clinical DDI study results, the mean DDI predictions from the final model slightly underestimated the clinically observed values. This may be attributed to the multitude of factors influencing TPN171 during co-administration, where inter-individual variability in physiological parameters and fluctuations in DDI-related parameters can alter DDI outcomes [28, 29], a phenomenon supported by subsequent population DDI simulations. Population DDI simulations for TPN171, incorporating variability in model parameters, demonstrated that the established PBPK model could satisfactorily predict the observed clinical DDI results.

Recent studies have substantially advanced the understanding of TPN171 pharmacokinetics and its susceptibility to metabolic interactions. He et al. [9] characterized the absorption, metabolism, and mass balance of TPN171 in healthy volunteers, establishing CYP3A4 as the primary metabolic pathway. Zhou et al. [30] further demonstrated that hepatic impairment significantly alters TPN171 exposure, warranting dose adjustment in this population. Most recently, Liang et al. [12] reported rifampicin significantly reduced TPN171 exposure in a clinical setting. While these findings collectively confirm that TPN171 is sensitive to CYP3A4 modulation, quantitative predictions across a broad spectrum of interacting drugs—particularly weak or moderate perpetrators—remain unavailable. Moreover, the combined effects of intrinsic (e.g., hepatic impairment) and extrinsic (e.g., CYP3A4 inhibition/induction) factors have not been systematically evaluated. Such information is critical for informing dose labeling and managing polypharmacy in real-world practice. The present study addresses these gaps by establishing and verifying a whole-body PBPK model for TPN171. Unlike conventional DDI assessments that rely on empirical classification, our model enables mechanism-based, quantitative prediction of exposure changes under varying degrees of CYP3A4 modulation. This approach not only recapitulates the clinical findings

reported by Liang et al. [12] under inducer co-administration, but also extends beyond tested scenarios to simulate untested inhibitor and inducer combinations. Collectively, these analyses position the current PBPK model as a complementary tool to existing clinical DDI data, offering regulatory and clinical stakeholders a quantitative basis for individualized dosing recommendations under diverse clinical contexts.

Applying the validated PBPK model to investigate DDIs between TPN171 and moderate or weak CYP3A4 perpetrators revealed that co-administration with moderate inhibitors (diltiazem 60 mg or fluconazole 400 mg) also resulted in substantial DDI effects. This underscores that the DDI outcome depends not only on the inhibitor's potency against CYP3A4 activity but also on factors such as the type of inhibition (e.g., diltiazem is a reversible and time-dependent inhibitor) [31], the dosing regimen (e.g., itraconazole was dosed for 4 days clinically vs. 8 days for diltiazem/fluconazole in simulations), and the hepatic exposure of the inhibitor. PBPK modeling demonstrated a markedly weaker DDI effect when TPN171 was co-administered with fluconazole 100 mg compared to the 400 mg dose, highlighting the significant impact of inhibitor exposure level on DDI magnitude.

The PBPK model was further utilized to explore the impact of dosing interval and dose adjustment on TPN171 exposure when co-administered with CYP3A4 perpetrators. Simulations indicated that exposure (AUC_{0-t}) increased by 1-to 2-fold when TPN171 was co-administered with strong or moderate CYP3A4 inhibitors. Model evaluation suggested that halving the TPN171 dose resulted in exposure levels under inhibition that were roughly equivalent to the exposure from twice that dose administered alone. Conversely, exposure decreased significantly with strong inducers. Even a 4-fold dose increase failed to achieve an AUC_{0-t} exceeding that of a quarter dose given alone, while the $C_{\max}$ was substantially elevated, potentially raising safety concerns due to sharp peak concentration increases; thus, co-administration is not recommended. With moderate CYP3A4 inducers, TPN171 exposure decreased by approximately half. While doubling the dose effectively increased AUC_{0-t} , it also concurrently raised $C_{\max}$, potentially introducing safety risks; therefore, dose adjustments should be considered cautiously based on the administered dose. Co-administration with CYP3A4 weak inhibitors and inducers resulted in negligible changes in $C_{\max}$ and AUC_{0-t} , indicating no need for dose adjustment.

Author Contributions

G.T., T.C., L.L. and J.J. wrote the manuscript; G.T., T.C. and J.J. designed the research; G.T., Z.W., H.Y., J.X.J., Y.D. and J.P. performed the research; and G.T., T.C. and J.J. analyzed the data.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Simulated and observed plasma concentration-time profiles of TPN171 after 10mg TPN171 alone for cohort1 (itraconazole DDI study) (A) and 20mg TPN171 alone for cohort2 (rifampin DDI study) (B) in healthy volunteers. Curve a indicates that TPN171 undergoes rapid and complete dissolution in the gastrointestinal tract, followed by a slightly slower and incomplete permeation process (Curve b shows that P-gp efflux reduces F_a to approximately 93%). Curve c and Curve d represent the drug concentrations reaching the hepatic portal vein and systemic circulation, respectively, demonstrating that TPN171 undergoes moderate intestinal first-pass metabolism mediated by CYP3A4, resulting in an oral bioavailability of approximately 65%. Curve e reflects the content of the parent drug in urine, with calculated proportions of the parent drug accounting for 10.68% and 11.51% of the administered oral dose for the 10 mg and 20 mg doses, respectively. The overlapping blue plasma PK curve and clinically observed data points indicate that the PBPK model effectively characterizes the ADME processes of TPN171 in humans. **Figure S2:** Population simulation for plasma concentration–time profiles in healthy subjects following oral administration of 10mg and 20mg TPN171 ($n=12$). Purple data points represent clinically observed individual plasma concentrations; blue curves indicate the 90% confidence intervals for individual PK profiles in population simulations; green bands represent the 90% confidence intervals for the mean PK curve in population simulations. **Figure S3:** Simulated and observed plasma concentration-time profiles of TPN171 after single oral dosing of 10mg TPN171 alone and co-administration with 200mg itraconazole (A) and single oral dosing of 20mg TPN171 alone and co-administration with 600mg rifampin (B) in healthy volunteers. **Figure S4:** PBPK model-predicted (solid line) and observed (solid dots) plasma concentrations of TPN171:(A) 5mg, (B) 10mg, (C) 20mg, (D) 30mg in single ascending dose (SAD) study. **Figure S5:** PBPK model-predicted (solid line) and observed (solid dots) plasma concentrations of TPN171:(A) 5mg in day 1, (B) 5mg in day 9, (C) 10mg in day 1, (D) 10mg in day 9 in multiple ascending dose (MAD) study. **Figure S6:** PBPK model-predicted (solid line) and observed (solid dots) plasma concentrations of 10 mgTPN171:(A) fasted, (B) fed in food effect (FE) study. **Figure S7:** PBPK model-predicted (solid line) and observed (solid dots) plasma concentrations of 5 mgTPN171in PK studies in the elderly. **Figure S8:** Single simulation (1) and population simulation (2) of TPN171 for plasma concentration–time profiles after single oral dosing of 10mg TPN171 alone and co-administration with CYP3A4 modulators in healthy volunteers: 60mg diltiazem (A1,A2); 400mg fluconazole (B1,B2); 100mg fluconazole (C1,C2); 50mg fluvoxamine (D1,D2); 400mg efavirenz (E1,E2); 160mg zanubrutinib (F1,F2). The solid line (blue/purple: TPN171 alone; red/green: co administration with CYP3A4 modulators) and shaded area represent predicted mean plasma concentration and its 95% confidence intervals, respectively (12 virtual individuals). **Table S1:** TPN171 clinical study data used in PBPK model development and verification. **Table S2:** PBPK Model-Predicted vs. Observed Geometric Mean of C_{max} and AUC for TPN171 in healthy volunteers. **Table S3:** PBPK Model-Predicted vs. Observed Geometric Mean Ratios of C_{max} and AUC for TPN171 as a Victim Drugin DDI Studies. **Table S4:** Comparison of PK parameters of different clinical studies for simulations and observed data in healthy volunteers. **Data S1:** Supporting Information. **Data S2:** Supporting Information. **Data S3:** Supporting Information. **Data S4:** Supporting Information.